

Assignment I
(Due: 22 March, 2026)

Question 1:

非原胞(惯用立方晶胞)

(a) Fig. Q1 shows a non-primitive cell of a lead-sulfide (PbS) crystal structure. The PbS crystal structure is built upon a face-centered cubic lattice with a basis of two different kinds of atoms (one Pb atom and one S atom) attached to each lattice point. The lattice constant for PbS is $a = 6.65 \text{ \AA}$. Let the origin of coordinates ^{坐标原点} of an x - y - z axes system is located at the lower back corner of the PbS cell and the coordinate axes are run along the edges of the cell (z upward) as shown in Fig. Q1. The basis with Pb atom is located at coordinate $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ and S atom is located at coordinate $(0,0,0)$.

- (i) Calculate the Pb atomic concentration (number of Pb atoms per cm^3) in the PbS.
(ii) Indicate (120) plane on Fig. Q1 and calculate the number of S atoms per cm^2 on the plane.

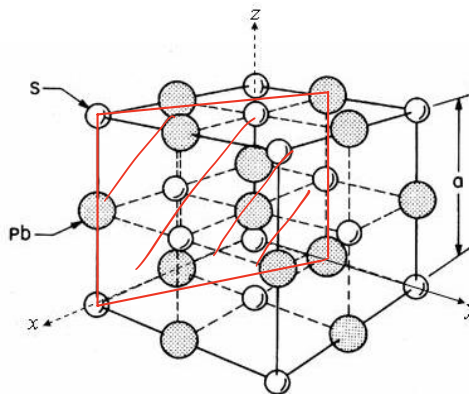


Fig. Q1

衍射强度

(b) Find the intensities of the diffraction $|S_{hkl}|^2$, for the primitive cell of PbS in terms of its atomic scattering factors of lead, f_{Pb} and sulfide f_S . What are the conditions that h, k, l have to satisfy to observe diffraction? Leave your answers in terms of h, k , and l .

(b) PbS的原胞包含2个原子 $\left\{ \begin{array}{l} \text{S原子位于}(0,0,0) \\ \text{Pb原子位于}(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}) \end{array} \right.$

原胞包含1个格点

这两个坐标

是基元中原子的相对位置

结构因子 $S_{hkl} = \sum f_j e^{-2\pi i(hx_j + ky_j + lz_j)}$

$$= f_S e^0 + f_{Pb} e^{-2\pi i(\frac{1}{2} + \frac{1}{2} + \frac{1}{2})}$$

$$= f_S + f_{Pb} e^{-i\pi(h+k+l)}$$

$$= f_S + f_{Pb} (-1)^{h+k+l}$$

$$\left\{ \begin{array}{l} \text{当 } h+k+l \text{ 为偶} \quad |S_{hkl}|^2 = (f_S + f_{Pb})^2 \\ \text{当 } h+k+l \text{ 为奇} \quad |S_{hkl}|^2 = (f_S - f_{Pb})^2 \end{array} \right.$$

(a)(i) 算Pb原子的体浓度

$$\begin{aligned} \text{原子数} \quad S &= 8 \times \frac{1}{8} + 6 \times \frac{1}{2} = 4 \\ P_b &= 12 \times \frac{1}{4} + 1 = 4 \end{aligned}$$

$$\text{单胞体积 } V = a^3 = (6.65 \times 10^{-8} \text{ cm})^3 \approx 2.941 \times 10^{-22} \text{ cm}^3$$

$$n_{Pb} = \frac{P_b \text{原子数}}{V} = \frac{4}{2.941 \times 10^{-22} \text{ cm}^3} \approx 1.36 \times 10^{22} \text{ atoms/cm}^3$$

(ii) 算S原子的面密度

$$\text{有效S原子数} = N_S = 2 \times \frac{1}{4} + 1 \times \frac{1}{2} = 1$$

$$\text{平面面积} = A = a \times \frac{\sqrt{2}}{2} a = \frac{\sqrt{2}}{2} a^2$$

$$\text{面密度 } \sigma_S = \frac{N_S}{A} = \frac{1}{\frac{\sqrt{2}}{2} a^2} \approx 2.02 \times 10^{14} \text{ atoms/cm}^2$$

原胞

$\Rightarrow$ 衍射条件:

① 对原胞指数: $\because f_Pb \neq f_S$

$$\forall h, k, l \quad |S_{hkl}|^2 \text{ 均不为 } 0$$

$\therefore \forall h, k, l$ 都能观察到衍射峰

② 对常规单胞指数: 需满足FCC消光规律

即 h, k, l 需全奇或全偶

Question 2:

A student has made an X-ray diffraction measurement on a sample. He obtained the first order diffraction peaks for the sample at the following $2\theta = 26.58^\circ, 37.94^\circ, 46.92^\circ, 54.74^\circ, 61.86^\circ, 68.53^\circ$ and 81.1° in ascending order, respectively. The X-ray wavelength used is $\lambda = 1.54 \text{ \AA}$. It is known that the sample has a cubic structure.

立方结构类型

(i) Determine the cubic structure type of the sample (i.e. simple cubic, body-centered cubic or face-centered cubic). Show your steps in arriving at the answer.

原理衍射面 (晶格常数)

(ii) List the first three allowed principle diffraction planes.

Given the lattice constant is 0.3350 nm.

(i)	序号	2θ	θ	$\sin^2\theta_i$	$\sin^2\theta_i/\sin^2\theta_1$	$\frac{4a^2\sin^2\theta}{\lambda^2} = h^2+k^2+l^2$	$\{hkl\}$
	1	26.58°	13.29°	0.05284	1	1	$\{100\}$
	2	37.94°	18.97°	0.10567	1.9996	2	$\{110\}$
	3	46.92°	23.46°	0.15849	2.9992	3	$\{111\}$
	4	54.74°	27.37°	0.21131	4	4	$\{200\}$
	5	61.86°	30.93°	0.26418	5	5	$\{210\}$
	6	68.53°	34.265°	0.31700	6	6	$\{211\}$
	7	81.1°	40.55°	0.42265	8	8	$\{220\}$

立方晶系中3整数平方和不可能为7

$\Rightarrow$ 为简单立方

(ii)	S=1	$\{100\}$ 晶面族	(100) 晶面
	S=2	$\{110\}$ 晶面族	(110) 晶面
	S=3	$\{111\}$ 晶面族	(111) 晶面

Bragg's Law: $n\lambda = 2d_{hkl}\sin\theta$

立方晶系晶面间距: $d_{hkl} = \frac{a}{\sqrt{h^2+k^2+l^2}}$

$\Rightarrow \sin^2\theta = \frac{n^2\lambda^2}{4a^2}(h^2+k^2+l^2)$

对同一实验相同

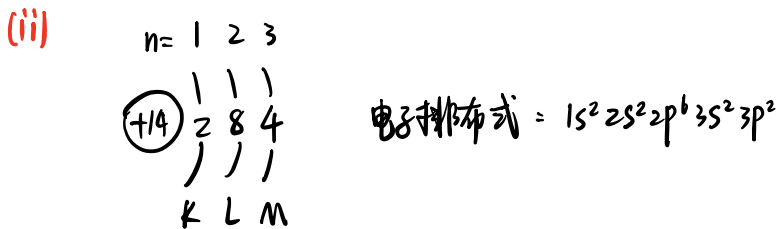
S 序列

- 简单立方 SC = 1, 2, 3, 4, 5, 6, 8... (所有 hkl)
- 体心立方 BCC = h+k+l 为偶, 2, 4, 6, 8, 10...
- 面心立方 FCC = h, k, l 全奇或全偶, 3, 4, 8, 11, 12...

Question 3:

- (i) The density of silicon is 2.33 g/cm^3 , and its relative atomic mass (RAM – mass in g of 1 mol, 6.022×10^{23} atoms per mol) is 28.09. Calculate the number of silicon atoms per cm^3 .
- (ii) The atomic number Z of Si is 14. Work out the electron configuration of silicon and determine the number of valence electrons per cm^3 of silicon.

(i) 单位体积内原子数 $n = \frac{\rho \times N_A}{M} = \frac{2.33 \text{ g/cm}^3 \times 6.022 \times 10^{23} \text{ atoms/mol}}{28.09 \text{ g/mol}} \approx 4.995 \times 10^{22} \text{ atoms/cm}^3$



单位体积内价电子数 $n_{\text{valence}} = 4 \text{ electrons/atom} \times 4.995 \times 10^{22} \text{ atoms/cm}^3 \approx 1.998 \times 10^{23} \text{ electrons/cm}^3$